TUDOR ȘERBAN

Senior Software Engineer | C++ Systems Architect | AI-assisted software development

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PROFESSIONAL PROFILE

Senior software engineer with more than 15 years of experience across algorithmic trading, game development, remote-access software and real-time industrial simulation. Strong modern C++ background, hands-on architecture and end-to-end implementation experience, and an academic foundation in intelligent and multi-agent systems. Currently designing a configurable C++20 trading platform while developing structured AI-agent-assisted workflows for planning, implementation, test-driven development and code review. Used to working independently, discussing requirements with domain specialists, and turning prototypes into working software.

TECHNICAL EXPERTISE

Languages & platforms: C++20, Python, Unreal Engine 5, Linux, macOS; earlier Visual C++, Unity and mobile platforms

Architecture & domains: Modular systems, automated trading, historical backtesting, game AI, real-time simulation, shared libraries, remote-access software

AI-assisted development: Cursor, specialized agent workflows, context separation, task planning, AI-assisted TDD, review and functional testing

Research & optimization: Multi-agent systems, genetic algorithms, parameter optimization, Bayesian optimization (Optuna), Ray Tune

Quality & operations: CMake, GoogleTest, OpenTelemetry, Docker, Kubernetes, WebSockets, REST APIs

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Equinox AI / *Jan 2025 – April 2026*

- Architected and implemented the core of a C++20 automated trading system supporting configurable strategies and evaluation against historical market data.
- Built the platform's core architecture and functionality in a two-person development team, taking primary responsibility for implementation and technical decisions.
- Set up a workflow using specialized AI agents in Cursor with distinct planning, implementation, TDD, code review and functional testing responsibilities; kept tasks and context separate so changes could be reviewed and tested.
- Developed a simplified Python trading prototype incorporating ML-based optimization of strategy parameters for experimentation.
- Used CMake, GoogleTest and OpenTelemetry; worked with Docker and Kubernetes in macOS and Linux environments.
- Collaborated with financial specialists on trading requirements and with infrastructure colleagues on deployment. The trading system is being deployed to cryptocurrency markets.

Independent Research & Development / *Dec 2022 – Dec 2024*

- Pursued self-directed research and prototypes in trading automation, agent-based game design and adaptive systems during a career break that also included parenting.
- Explored genetic algorithms and agent behaviors, including experimentation with AI-assisted design workflows.
- Studied financial markets and Python-based approaches to trading-system experimentation.

Senior Software Engineer | Dream Primer / *2022*

- Developed AI and gameplay systems for an Unreal Engine 5 MMORPG pilot, including dynamic monster behavior.
- Integrated gameplay AI with networking and user-interface components.

Senior Software Engineer | Gameloft Romania / *2017 – 2021*

- Developed and maintained shared libraries for multiplayer, social and advertising functionality used across game projects, including the Asphalt franchise.
- Worked on scalable, reusable components and integration across teams and projects, using Visual C++, Unity and Python.

Scrum Master & Software Developer | Netop Tech / *2012 – 2017*

- Combined software development and Scrum Master responsibilities for a team of approximately 20.
- Worked on cloud-enabled remote-access functionality in established Visual C++ products, including WebSocket and REST API integrations.

Software Engineer | Namco Bandai / Mikon Systems / *2004 – 2012*

- Designed and implemented real-time industrial process and power-system simulators, modeling electrical circuitry, sensors and actuators.
- Worked on mobile game porting and development with ARM compiler, BREW SDK and J2ME technologies.

EDUCATION

Master's Degree in Intelligent Systems | University POLITEHNICA of Bucharest

Specialization: Multi-Agent Systems. Academic focus included adaptive behavior, control algorithms, robotics and cooperative robot teams.

Thesis/research topic: simulation of a robot soccer team using a behavior-based approach.

SELECTED RESEARCH PUBLICATIONS

- "Behavioral Diversity in Cooperative Multi-Robot Tasks" | RAAD 2005.
- "A Method for the Design of Adaptive Behavior-Based Cooperative Robot Teams."
- "Behavior-Based Multi-Robot Systems Control" | ISOM Workshop, 2005.

SELECTED PROJECTS

AI-assisted development workflows / *Private*

Custom engineering workflows using specialized AI agents for technical planning, implementation, TDD, review and functional validation. Applied to modern C++ development and explored in game-development contexts.

Emergent-Behavior Game Design / *Independent R&D*

Agent-based game-design research exploring goal-seeking behavior and genetic-algorithm optimization, drawing on earlier work with multi-agent systems.

Note: Project code is private. Additional technical details can be discussed during interviews.